

Day 24. Skewed datasets

Error metrics for skewed datasets

Rare disease classification example

Train classifier $f_{\vec{w},b}(\vec{x})$ ($y = 1$ if disease present,
 $y = 0$ otherwise)

Find that you've got 1% error on test set
(99% correct diagnoses)

Only 0.5% of patients have the disease

99.5% accuracy, 0.5% error

```
print("y=0")
```

Precision/recall

$y = 1$ in presence of rare class we want to detect.

Actual Class		1	0
Predicted Class	1	True positive 15	False positive 5
	0	False negative 10	True negative 70
		↓ 25	↓ 75

Precision:

(of all patients where we predicted $y = 1$, what fraction actually have the rare disease?)

$$\frac{\text{True positives}}{\text{\#predicted positive}} = \frac{\text{True positives}}{\text{True pos} + \text{False pos}} = \frac{15}{15 + 5} = 0.75$$

Recall:

(of all patients that actually have the rare disease, what fraction did we correctly detect as having it?)

$$\frac{\text{True positives}}{\text{\#actual positive}} = \frac{\text{True positives}}{\text{True pos} + \text{False neg}} = \frac{15}{15 + 10} = 0.6$$

Trading off precision and recall

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Logistic regression: $0 < f_{\vec{w},b}(\vec{x}) < 1$

- Predict 1 if $f_{\vec{w},b}(\vec{x}) \geq 0.5$ ~~0.7~~ ~~0.9~~ 0.3
- Predict 0 if $f_{\vec{w},b}(\vec{x}) < 0.5$ ~~0.7~~ ~~0.9~~ 0.3

$$\text{precision} = \frac{\text{true positives}}{\text{total predicted positive}}$$

$$\text{recall} = \frac{\text{true positives}}{\text{total actual positive}}$$

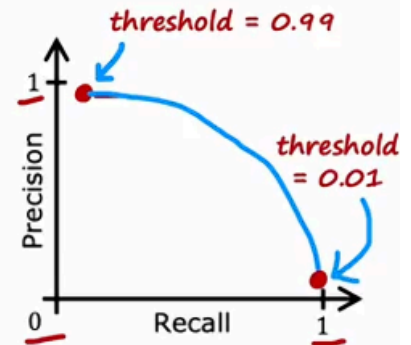
Suppose we want to predict $y = 1$ (rare disease) only if very confident.

higher precision, lower recall

Suppose we want to avoid missing too many case of rare disease (when in doubt predict $y = 1$)

lower precision, higher recall

More generally predict 1 if: $f_{\vec{w},b}(\vec{x}) \geq \text{threshold}$.



F1 score

How to compare precision/recall numbers?

	Precision (P)	Recall (R)	Average	F ₁ score
Algorithm 1	0.5	0.4	0.45	0.444
Algorithm 2	0.7	0.1	0.4	0.175
Algorithm 3	0.02	1.0	0.501	0.0392

`print("y=1")`

~~Average = $\frac{P+R}{2}$~~

$$F_1 \text{ score} = \frac{1}{\frac{1}{2} \left(\frac{1}{P} + \frac{1}{R} \right)} = 2 \frac{PR}{P+R}$$